

Also, because of the way overlapping periods are formed with historical simulations, the middle part of the historical record plays a disproportionately important role in the analysis. With data since 1926 and for thirty-year retirement durations, 1926 appears in one rolling historical simulation, while 1927 appears in two (for the 1926 and 1927 retirees). This pattern continues until 1955, which appears in thirty simulations (the last year for the 1926 retiree through the first year for the 1955 retiree). The years 1955 through 1987 all appear in thirty simulated retirements, with 1988 appearing in twenty-nine simulations, through 2016, which only appears in one simulation (as the final year of retirement for the 1987 retiree). This overweighted portion of the data (1955–1987) coincides with a bear market for bonds.

In contrast, Monte Carlo simulations treat each data point as equally important. This has implications for portfolio success rates with different asset allocations. Essentially, bonds will have more opportunity to shine with Monte Carlo. In addition, with many more simulations we can expect some scenarios where a particular spending strategy will not work even when it always worked in the same historical data that drives the simulations. Monte Carlo simulations of the 4 percent rule based on the same underlying data as historical simulations tend to show greater relative success for bond-heavy strategies, less relative success for stock-heavy strategies, and lower optimal stock allocations.

Exhibit 4.6 compares the portfolio success rates for varying asset allocations when using a 4 percent withdrawal rate over thirty-year periods. Withdrawals of 4 percent did not fail in the historical period for stock allocations between 40 and 70 percent. Bond-heavy portfolios experienced much lower success rates, though, with a bonds-only portfolio providing success in 44 percent of the historical simulations. With Monte Carlo simulations based on the same historical data, retirees would be encouraged to hold some stocks, but success rates of over 90 percent are possible with stock allocations of only 20 percent. The highest success rates occur in the range between 30 and 60 percent stocks.

Exhibit 4.6

Portfolio Success Rates for a 4% Withdrawal Rate